

Convert Your Old DVD Player With USB to a DVP



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A digital versatile disk (DVD) player is an optoelectronic decoding device that sends a red laser beam to the DVD disk that is run at a certain speed. Playing a DVD continuously may heat up the player and cause problems with the optical pick-up assembly section, which, in turn, may stop the DVD player from working completely. One way to solve this problem is described in this article. The DVD player should have a pre-installed USB port, though. We can discard the CD/DVD drive section and use the USB port to convert it into a USB

digital video player (DVP).

This article describes how to convert an old Sony DVD player (model CD/DVD DVP-SR660P) into a DVP. Author's prototype of the USB DVP is shown in Fig. 1. A USB DVP has three main PCBs: power supply, audio/video output section with 5.1 channel audio output and front panel with a 5-digit, 7-segment display, and a TSOP-1738 module [infrared (IR) receiver], a few tactile switches and a USB 2.0 host.

The power supply PCB provides the power for the DVP player. It has all necessary components like resistors and capacitors. It also has an AC-in connector for connecting the DVP to mains supply.

The front panel PCB, as shown in Fig. 2, has two cables, one for the 7-segment display and the other for the USB host-controller cable. Tactile switches are used for functions like on/off, forward/backward, next/previous, pause/resume, stop and the like. The 7-segment display is used to display the operations performed by the user and the USB host is used to connect a USB drive.

The AV (audio/video) controller board, as shown in Fig. 3, is used to control the audio and video functions of the DVP.

AV-out RCA and 5.1-channel audio-out cables are connected to the AV controller board. The board has composite video-out (yellow) RCA cable, component video-out cables (Y: green component, PR: red component and PB: blue component).

Construction and testing

The final prototype of the USB DVP player is shown in Fig. 4. Make all connections as per the figure.

Step-by-step construction and operation is given below:

1. Open the DVD player cover with the help of a screwdriver.

2. Remove all three PCBs, that is, power PCB, front panel PCB and AV controller board, from the DVD player carefully. Remove all connections from each of these boards without touching any capacitor, which may be highly charged.

3. Place the three PCBs on a rectangular or square insulating enclosure. Here, we have used a plastic box.

4. Place the AV controller board at the rear side of the box, so that connections would be easy for connecting RCA and audio-out cables.

5. Place the power PCB near the AV controller board (Fig. 4) so that connectors running from the power PCB board reach and sit securely on the receptacle of the AV controller board.

6. Place the front panel PCB at the front of the box.

7. Connect the AC mains cable in the AC-in connector on the power PCB. All connections should be made properly, as any wrong connection may damage the boards.

8. Connect the composite video-out (yellow) RCA cable; in absence of that cable, you may connect the component video-out cable (Y: green



Fig. 1: Author's prototype of the USB DVP



Fig. 2: Front panel PCB (5-digit display, IR receiver and USB host controller)